

Laporan
ISB - 310 Sistem Informasi Berbasis Web (Praktikum)

Oleh:

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PROGRAM STUDI SISTEM INFORMASI
FAKULTAS TEKNOLOGI INDUSTRI
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2026

1. Soal 1

Nama Program	: Sistem informasi Rental mobil Land Rover v5
Kode Program	<pre><?php use Illuminate\Database\Migrations\Migration; use Illuminate\Database\Schema\Blueprint; use Illuminate\Support\Facades\Schema; return new class extends Migration { /** * Run the migrations. */ public function up(): void { Schema::create('categories', function (Blueprint \$table) { \$table->id('category_id')->primaryKey(); \$table->string('category_name'); \$table->timestamps(); }); } /** * Reverse the migrations. */ public function down(): void { Schema::dropIfExists('categories'); } }; <?php use Illuminate\Database\Migrations\Migration; use Illuminate\Database\Schema\Blueprint; use Illuminate\Support\Facades\Schema; return new class extends Migration { /**</pre>

```
        * Run the migrations.
        */
    public function up(): void
    {
        Schema::create('products', function (Blueprint $table) {
            $table->id('product_id')->primarykey();
            $table->unsignedBigInteger('category_id');
            $table->foreign('category_id')->
>references('category_id')->on('categories')->onDelete('cascade');
            $table->unsignedBigInteger('brand_id');
            $table->foreign('brand_id')->references('brand_id')->
>on('brands')->onDelete('cascade');
            $table->string('product_name');
            $table->integer('product_price');
            $table->integer('product_stock');
            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('products');
    }
};
```

```
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('brand', function (Blueprint $table) {
```

```
        $table->id('brand_id')->primarykey();  
        $table->string('brand_name');  
        $table->timestamps();  
    });  
}
```

```
/**  
 * Reverse the migrations.  
 */  
public function down(): void  
{  
    Schema::dropIfExists('brands');  
}  
};
```

```
<?php
```

```
namespace App\Models;
```

```
use Illuminate\Database\Eloquent\Model;
```

```
class product extends Model
```

```
{
```

```
    protected $table = 'products';
```

```
    protected $primaryKey = 'product_id';
```

```
    public $timestamps = false;
```

```
    protected $fillable = [  
        'product_id',  
        'category_id',  
        'brand_id',  
        'product_name',  
        'product_price',  
        'product_stock'  
    ];
```

```
    public function category()  
{  
        return $this->belongsTo(category::class, 'category_id',  
'category_id');  
}
```

```
    public function brand()
```

```
{  
    return $this->belongsTo(category::class, 'brand_id',  
    'brand_id');  
}  
}
```

```
<?php  
  
namespace App\Models;  
  
use Illuminate\Database\Eloquent\Model;  
  
class category extends Model  
{  
    protected $table = 'categories';  
    protected $primaryKey = 'category_id';  
    public $timestamps = false;  
    protected $fillable = [  
        'category_id',  
        'category_name'  
    ];  
  
    public function products()  
    {  
        return $this->hasMany(product::class, 'category_id',  
        'category_id');  
    }  
}
```

```
<?php  
  
namespace App\Models;  
  
use Illuminate\Database\Eloquent\Model;  
  
class brand extends Model  
{  
    protected $table = 'brand';  
    protected $primaryKey = 'brand_id';  
    public $timestamps = false;
```

```
protected $fillable = [
    'brand_id',
    'brand_name'
];

public function products()
{
    return $this->hasMany(product::class, 'brand_id',
'brand_id');
}
```

Z:\laragon\www\rental-landRover>php artisan db:seed

INFO Seeding database.

INFO Controller [Z:\laragon\www\rental-landRover\app\Http\Controllers\ProductController.php] created successfully.

Z:\laragon\www\rental-landRover>php artisan make:view product

INFO View [Z:\laragon\www\rental-landRover\resources\views\product.blade.php] created successfully.

rental-landrover.products: 6 rows total (approximately)

>> Next

Show all

Sorting

Columns (8)

product_id	category_id	brand_id	product_name	product_price	product_stock	created_at	updated_at
1	1	1	Range Rover Vogue	35.000	5	2026-04-08 02:29:43	2026-04-08 02:29:43
2	2	2	Land Rover Defender 110	28.000	12	2026-04-08 02:29:43	2026-04-08 02:29:43
3	3	1	Range Rover Evoque	15.000	8	2026-04-08 02:29:43	2026-04-08 02:29:43
4	1	3	Range Rover Velar	21.000	4	2026-04-08 02:29:43	2026-04-08 02:29:43
5	2	4	Discovery Sport	18.000	7	2026-04-08 02:29:43	2026-04-08 02:29:43
6	4	3	Land Rover Series III	75.000	2	2026-04-08 02:29:43	2026-04-08 02:29:43

brand_id	brand_name	created_at	updated_at
1	Ingenium Engine	2026-04-08 02:29:43	2026-04-08 02:29:43
2	BMW V8 Power	2026-04-08 02:29:43	2026-04-08 02:29:43
3	Ford EcoBoost	2026-04-08 02:29:43	2026-04-08 02:29:43
4	ZF Transmission	2026-04-08 02:29:43	2026-04-08 02:29:43

category_id	category_name	created_at	updated_at
1	Luxury SUV	2026-04-08 02:29:43	2026-04-08 02:29:43
2	Off-Road Specialist	2026-04-08 02:29:43	2026-04-08 02:29:43
3	Compact SUV	2026-04-08 02:29:43	2026-04-08 02:29:43
4	Classic Series	2026-04-08 02:29:43	2026-04-08 02:29:43

<?php

namespace App\Http\Controllers;

```
use App\Models\brand;
use App\Models\category;
use App\Models\product;
use Illuminate\Http\Request;

class ProductController extends Controller
{
    /**
     * Display a listing of the resource.
     */
    public function index()
    {
        $products = product::with(['category', 'brand'])
            ->orderBy('product_name')
            ->get();

        return view('product', compact('products'));
    }

    /**
     * Show the form for creating a new resource.
     */
    public function create()
    {
        $categories = category::orderBy('category_name')->get();
        $brands = brand::orderBy('brand_name')->get();

        return view('product-create', compact('categories',
'brands'));
    }

    /**
     * Store a newly created resource in storage.
     */
    public function store(Request $request)
    {
        $validated = $request->validate([
            'category_id' =>
'required|exists:categories,category_id',
            'brand_id' => 'required|exists:brands,brand_id',
            'product_name' => 'required|string|max:255',
            'product_price' => 'required|integer|min:0',
        ]);
    }
}
```

```
        'product_stock' => 'required|integer|min:0',
    ]);

    product::create($validated);

    return redirect()
        ->route('products')
        ->with('success', 'Produk berhasil ditambahkan.');
```

```
    }

    /**
     * Display the specified resource.
     */
    public function show(string $id)
    {
        //
    }

    /**
     * Show the form for editing the specified resource.
     */
    public function edit(string $id)
    {
        //
    }

    /**
     * Update the specified resource in storage.
     */
    public function update(Request $request, string $id)
    {
        //
    }

    /**
     * Remove the specified resource from storage.
     */
    public function destroy(string $id)
    {
        //
    }
}
```

```
<?php

namespace Database\Seeders;

use Illuminate\Database\Console\Seeds\WithoutModelEvents;
use Illuminate\Database\Seeder;
use Illuminate\Support\Facades\DB;
use Illuminate\Support\Carbon;

class DatabaseSeeder extends Seeder
{
    use WithoutModelEvents;

    public function run(): void
    {
        $now = Carbon::now();

        DB::table('categories')->insert([
            [
                'category_id' => 1,
                'category_name' => 'Luxury SUV',
                'created_at' => $now,
                'updated_at' => $now,
            ],
            [
                'category_id' => 2,
                'category_name' => 'Off-Road Specialist',
                'created_at' => $now,
                'updated_at' => $now,
            ],
            [
                'category_id' => 3,
                'category_name' => 'Compact SUV',
                'created_at' => $now,
                'updated_at' => $now,
            ],
            [
                'category_id' => 4,
                'category_name' => 'Classic Series',
                'created_at' => $now,
                'updated_at' => $now,
            ],
        ]);
    }
}
```

```
DB::table('brands')->insert([
    [
        'brand_id' => 1,
        'brand_name' => 'Ingenium Engine',
        'created_at' => $now,
        'updated_at' => $now,
    ],
    [
        'brand_id' => 2,
        'brand_name' => 'BMW V8 Power',
        'created_at' => $now,
        'updated_at' => $now,
    ],
    [
        'brand_id' => 3,
        'brand_name' => 'Ford EcoBoost',
        'created_at' => $now,
        'updated_at' => $now,
    ],
    [
        'brand_id' => 4,
        'brand_name' => 'ZF Transmission',
        'created_at' => $now,
        'updated_at' => $now,
    ],
]);

DB::table('products')->insert([
    [
        'product_id' => 1,
        'category_id' => 1,
        'brand_id' => 1,
        'product_name' => 'Range Rover Vogue',
        'product_price' => 35000,
        'product_stock' => 5,
        'created_at' => $now,
        'updated_at' => $now,
    ],
    [
        'product_id' => 2,
        'category_id' => 2,
        'brand_id' => 2,
```

```
'product_name' => 'Land Rover Defender 110',
'product_price' => 28000,
'product_stock' => 12,
'created_at' => $now,
'updated_at' => $now,
],
[
  'product_id' => 3,
  'category_id' => 3,
  'brand_id' => 1,
  'product_name' => 'Range Rover Evoque',
  'product_price' => 15000,
  'product_stock' => 8,
  'created_at' => $now,
  'updated_at' => $now,
],
[
  'product_id' => 4,
  'category_id' => 1,
  'brand_id' => 3,
  'product_name' => 'Range Rover Velar',
  'product_price' => 21000,
  'product_stock' => 4,
  'created_at' => $now,
  'updated_at' => $now,
],
[
  'product_id' => 5,
  'category_id' => 2,
  'brand_id' => 4,
  'product_name' => 'Discovery Sport',
  'product_price' => 18000,
  'product_stock' => 7,
  'created_at' => $now,
  'updated_at' => $now,
],
[
  'product_id' => 6,
  'category_id' => 4,
  'brand_id' => 3,
  'product_name' => 'Land Rover Series III',
  'product_price' => 75000,
  'product_stock' => 2,
```

The screenshot displays a web application for 'Land Rover Rental'. The header features the brand logo, navigation links (Home, Produk, Booking, Tamu), and user options (Light Mode, Login, Wishlist). The main banner promotes 'Sewa Land Rover Untuk Petualangan Anda' (Rent Land Rover for Your Adventure) with a background image of a white SUV on a scenic road. Below the banner, a section titled 'Armada Kami' (Our Fleet) lists six vehicles in a grid:

- Range Rover 2005:** Kemewahan klasik dengan performa off-road mumpuni. Interior kulit, mesin V8. Stok: 5 unit. Price: Rp 1.500.000/hari.
- Range Rover 2007:** Model facelift dengan teknologi lebih modern, tetap elegan dan tangguh. Stok: 4 unit. Price: Rp 1.700.000/hari.
- Range Rover Sport 2005:** Dinamis dan sporty, cocok untuk perjalanan berkecepatan tinggi dengan gaya. Stok: 6 unit. Price: Rp 1.600.000/hari.
- Discovery 3:** Ruang lega, mampu membawa keluarga besar dengan kenyamanan terbaik.
- Discovery 4:** Perpaduan kemewahan dan teknologi canggih, siap menemani petualangan Anda.
- Land Rover Defender 2007:** Ikon off-road sejati, kokoh dan tangguh di segala medan. Pilihan tepat untuk petualangan ekstrem.

The footer contains the company name 'Land Rover Rental', a brief description of the service, a menu (Home, Booking), social media links, and a copyright notice: '© 2025 Land Rover Rental. All rights reserved.'

Land Rover Rental

HomeProdukBookingTamu

Daftar Produk Land Rover

Semua produk di bawah ini ditampilkan langsung dari database beserta kategori, brand, harga, dan stok saat ini.

Booking Sekarang

Produk Tersedia

Tambah Produk6 produk



Off-Road SpecialistZF Transmission

Discovery Sport

ID Produk: 5

Harga: **Rp 18.000**

Stok: 7 unit

Sewa Produk Ini



Off-Road SpecialistBMW V8 Power

Land Rover Defender 110

ID Produk: 2

Harga: **Rp 28.000**

Stok: 12 unit

Sewa Produk Ini



Classic SeriesFord EcoBoost

Land Rover Series III

ID Produk: 6

Harga: **Rp 75.000**

Stok: 2 unit

Sewa Produk Ini



Compact SUVIngenium Engine

Range Rover Evoque

ID Produk: 3

Harga: **Rp 15.000**

Stok: 8 unit

Sewa Produk Ini



Luxury SUVFord EcoBoost

Range Rover Velar

ID Produk: 4

Harga: **Rp 21.000**

Stok: 4 unit

Sewa Produk Ini



Luxury SUVIngenium Engine

Range Rover Vogue

ID Produk: 1

Harga: **Rp 35.000**

Stok: 5 unit

Sewa Produk Ini

© 2025 Land Rover Rental. All rights reserved.
Kembali ke halaman utama

Land Rover Rental

Kembali ke Produk

Tambah Produk Baru

Nama Produk

Kategori

Pilih kategori

Brand

Pilih brand

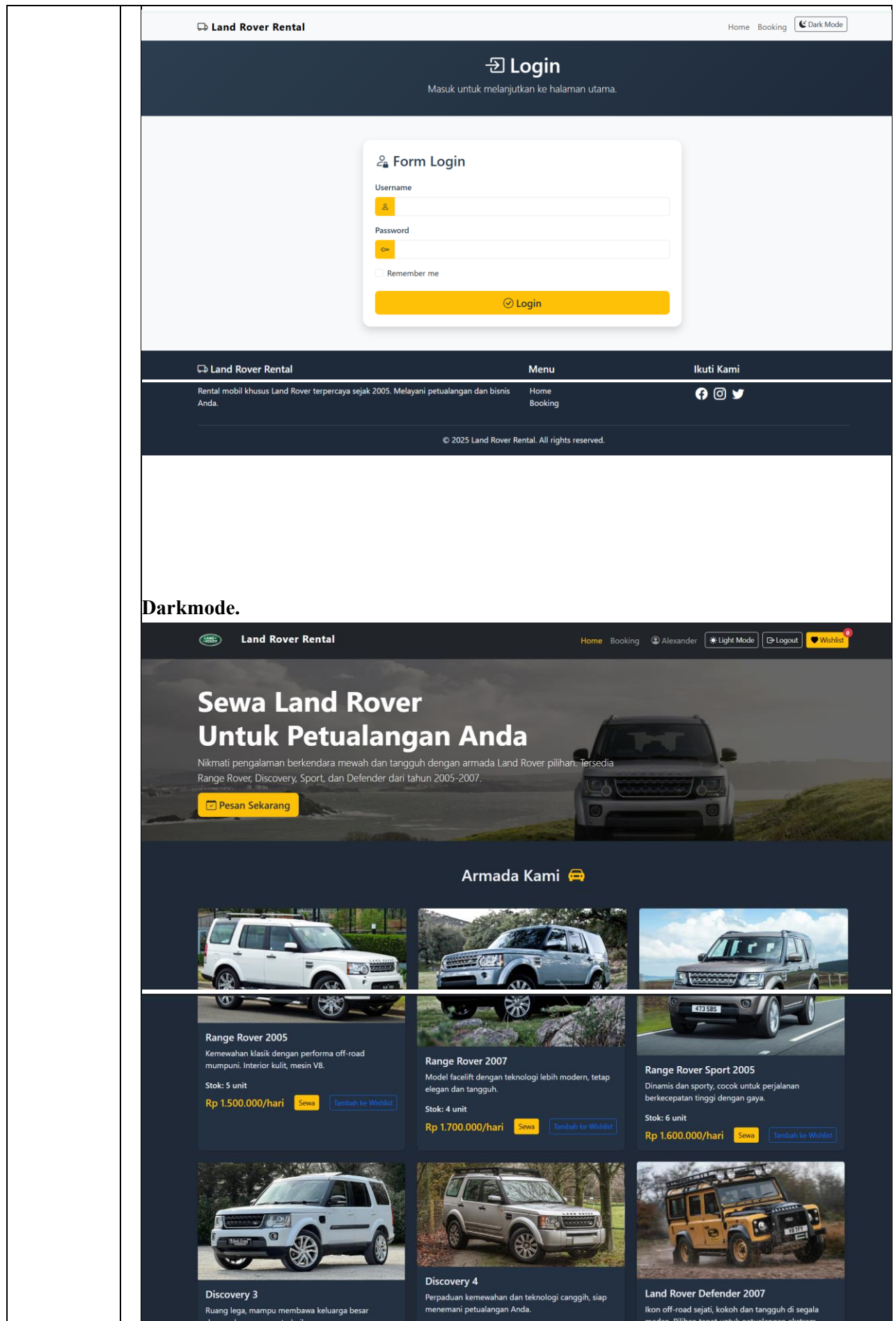
Harga

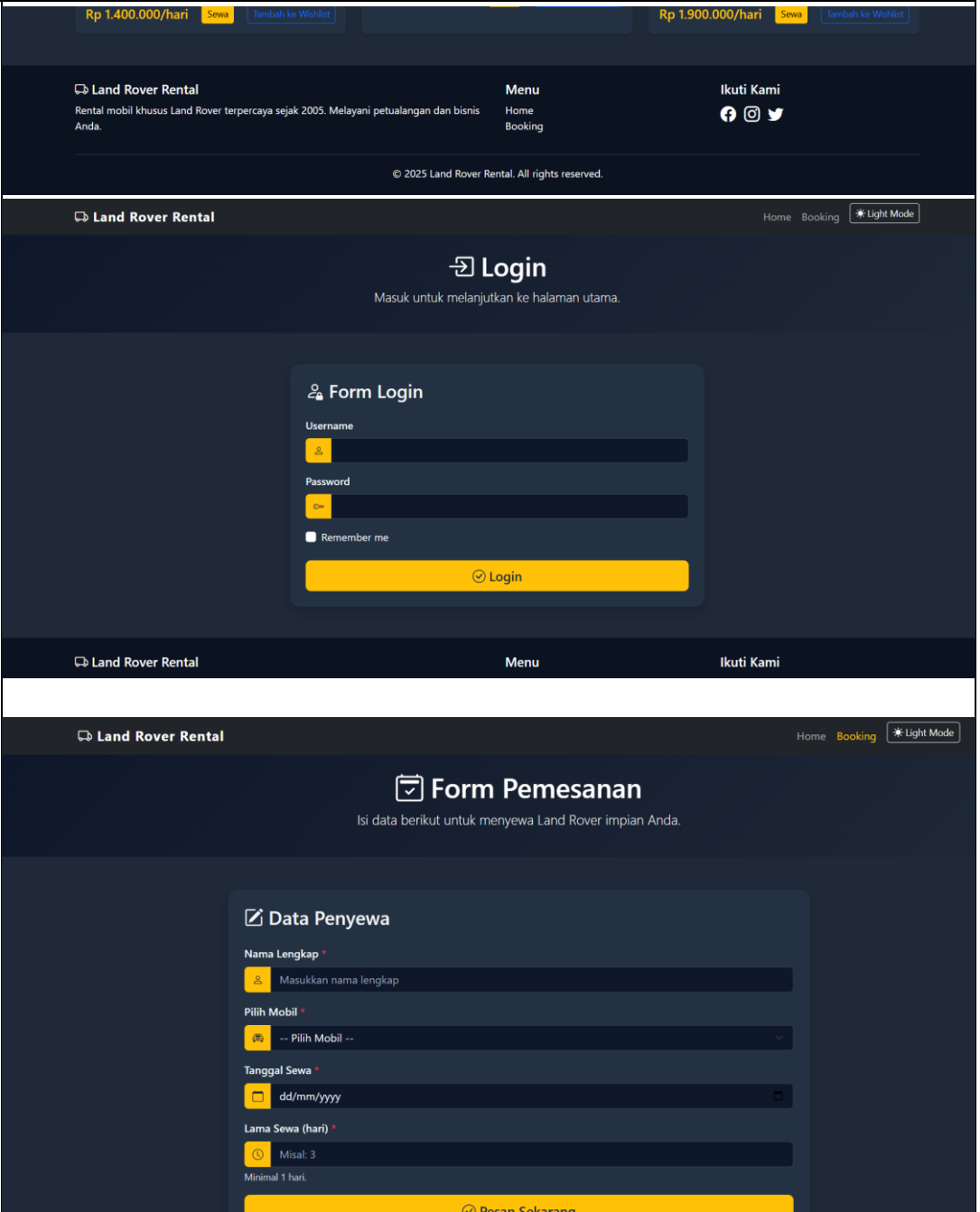
Stok

Simpan Produk

Batal

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| | |
|--------------------------|--|
| |  <p>The image shows two screenshots of a web application for 'Land Rover Rental'. The top screenshot is the 'Login' page, which features a 'Form Login' with fields for 'Username' and 'Password', a 'Remember me' checkbox, and a 'Login' button. The bottom screenshot is the 'Form Pemesanan' (Booking) page, which features a 'Data Penyewa' (Renter Data) form with fields for 'Nama Lengkap' (Full Name), 'Pilih Mobil' (Select Car), 'Tanggal Sewa' (Rental Date), and 'Lama Sewa (hari)' (Rental Duration). Both pages have a dark blue header with the company name, navigation links, and social media icons.</p> |
| <p>Penjelasan</p> | <p>Perubahan dimulai dari layer database, yaitu migration dan seeder. Pada migration, struktur tabel categories, brands, dan products dirapikan agar saling terhubung dengan benar: products sekarang memiliki foreign key category_id dan brand_id yang mengarah ke tabel categories dan brands, sementara penamaan tabel dan urutan migration juga dibetulkan supaya tidak terjadi error relasi saat migrate. Selain itu, tipe dan struktur kolom disesuaikan dengan data yang akan disimpan, lalu DatabaseSeeder diperbaiki agar field yang diinsert sesuai dengan skema tabel, termasuk penambahan brand_id pada data produk dan pengisian created_at serta updated_at.</p> <p>Setelah struktur database benar, bagian model Eloquent disesuaikan agar Laravel bisa membaca relasi antar tabel. Model product dihubungkan ke model category dan brand melalui relasi belongsTo, sedangkan model category dan brand memiliki relasi hasMany ke produk. Beberapa kesalahan awal juga diperbaiki, seperti model brand yang sebelumnya masih menunjuk ke nama tabel yang salah, serta relasi brand() pada model product yang sempat mengarah ke class yang tidak</p> |

| | |
|--|--|
| | <p>sesuai. Dengan perbaikan ini, data produk bisa diambil sekaligus bersama data kategori dan brand-nya menggunakan query Eloquent.</p> <p>Pada sisi controller dan view, ProductController@index dibuat untuk mengambil seluruh data produk dari database dengan eager loading relasi category dan brand, lalu mengirimkannya ke halaman product.blade.php. Di halaman tersebut, data tidak lagi ditulis statis, tetapi ditampilkan dengan perulangan @forelse langsung dari database, sehingga nama produk, kategori, brand, harga, dan stok selalu mengikuti isi tabel. Dari halaman indeks.blade.php, juga ditambahkan tombol dan navigasi menuju halaman produk, sehingga alur aplikasi menjadi lengkap: data disimpan di database, dibaca lewat model dan controller, lalu ditampilkan secara dinamis di halaman produk.</p> |
|--|--|